Problem 1

$$\mathbf{r}(t) = \left\langle t^2,\ \frac{t^3 - 1}{t - 1},\ t + 1 \right\rangle, \qquad t \neq 1.$$

1. Find the domain of $\mathbf{r}$, and evaluate $\lim_{t \to 1} \mathbf{r}(t)$.
2. Define $\mathbf{r}(1)$ so that $\mathbf{r}$ becomes continuous. Denote the continuous extension by $\widetilde{\mathbf{r}}$.
3. Let C be the space curve traced by $\widetilde{\mathbf{r}}$. Find two equations in x, y, z that describe C, with no parameter t.
4. Find all points of C whose x-coordinate is 4. Does C determine z as a function of x? Briefly explain.

Problem 2

$$\mathbf{r}(t) = \langle t^2,\ t^3 - 3t,\ 2t \rangle.$$

1. Let $P = \mathbf{r}(1)$. Find the speed, the unit tangent vector, and a tangent line at P.
2. Find the normal plane to the curve at P.
3. Find the other point $Q \neq P$ where the curve intersects this normal plane, and compute $P - Q$.
4. Compute the average rate of change of $\mathbf{r}(t)$ from $t = -3$ to $t = 1$. Explain why this vector is orthogonal to the tangent direction at P.

Problem 3

$$\mathbf{r}(t) = \big\langle 3\cos(t^2-1),\ 3\sin(t^2-1),\ 4(t^2-1)\big\rangle, \qquad 1 \le t \le 2.$$

1. Find the speed $\|\mathbf{r}'(t)\|$ and the total arc length of the curve.
2. Find the arc length function $s(t)$, measured from $t = 1$. Then find the point P for which $s = 5\pi/2$.
3. Find the unit tangent vector and the curvature at P.
4. Find **N** and **B** at P. Then find an equation of the osculating plane at P.